Rule Bach or not?

Does rule injection make Bach-style chorale generation better?

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Introduction

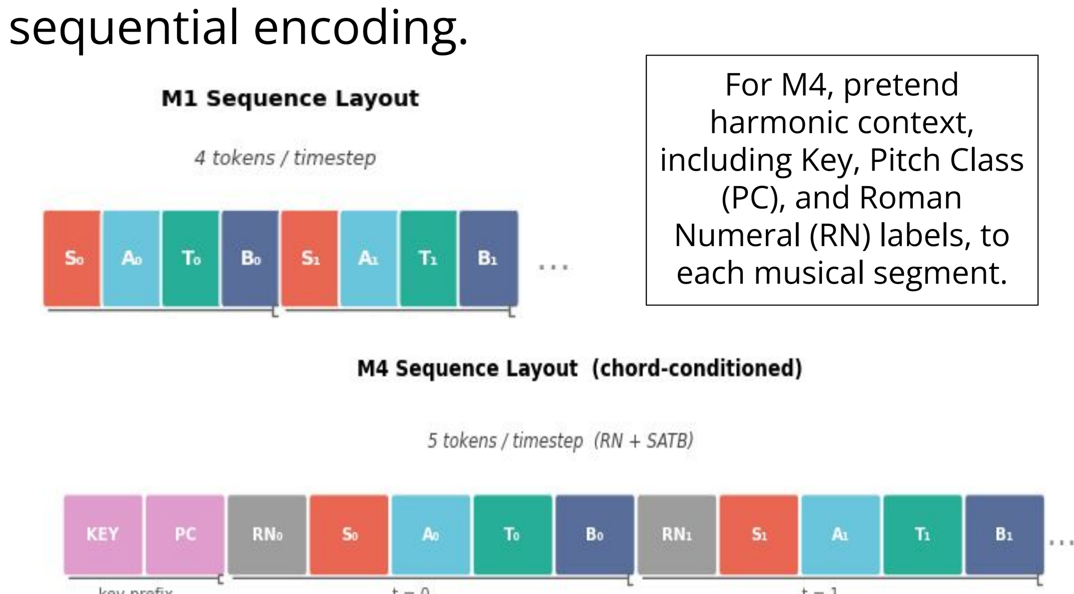
Most contemporary models for Bach chorale generation, such as Coconet, are purely data-driven, which results in frequent violations in fundamental music theory rules such as parallel fifths and octaves.

In this project, we study **four-part Bach chorale generation as an autoregressive sequence modeling problem with explicit music-theory rule injection**. Our goal is to test whether symbolic rule knowledge will improve the generation quality and where symbolic rule knowledge is most effective.

Dataset

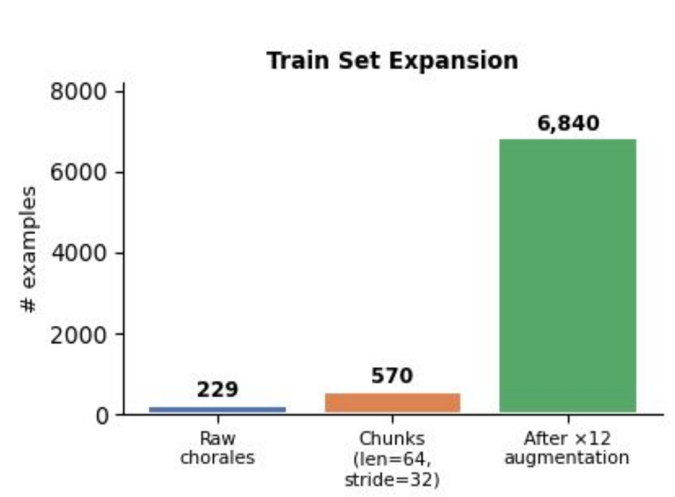
JSB Chorales dataset: 229 trains, 76 validation, and 77 test samples.

Data Preprocess: All pieces are quantized to a 16th-note resolution and performed a structured sequential encoding.



The model learns a vocabulary of 103 unique tokens (M4), encompassing both musical notes and theoretical metadata.

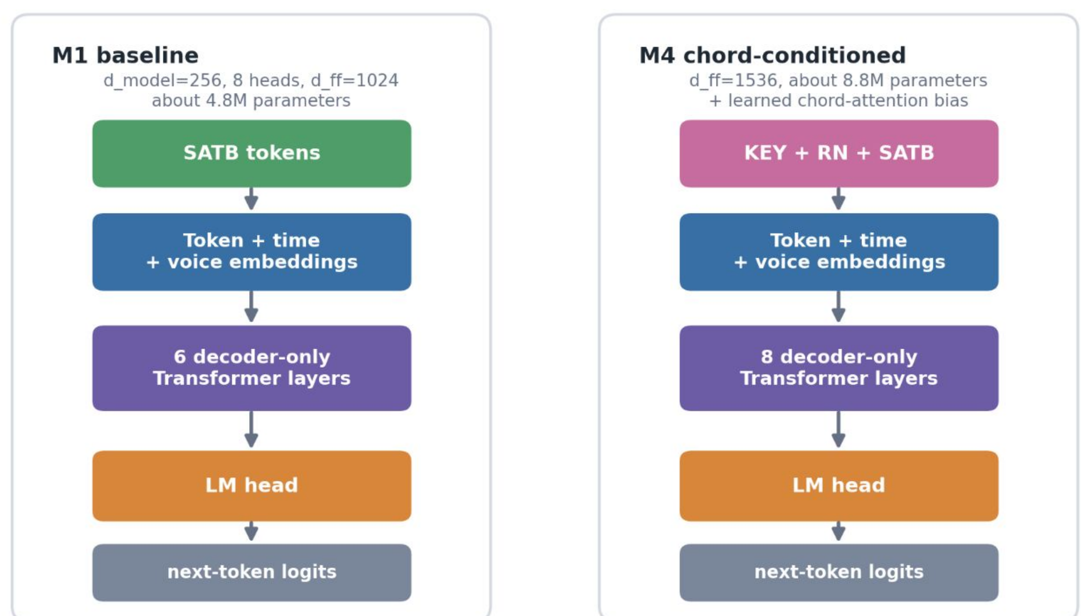
Data Augmentation: Each chorale was transposed into all 12 musical keys and long sequences were fragmented into 64 time steps.



Methodology

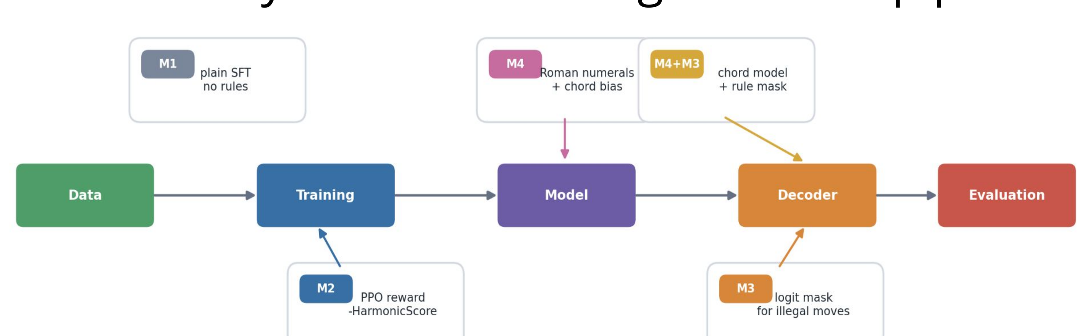
Model Architecture

We trained a decoder-only Transformer from scratch. The baseline model used 6 layers and learned embeddings for both time position and voice role. The chord-conditioned model extends the same autoregressive setup with a larger 8-layer Roman-numeral tokens and a learned chord-attention bias.



Experiment Conditions

We create 5 conditions to compare where music-theory rules enter the generation pipeline:



Rule Checker

All models are evaluated by generating chorales and measuring **HarmonicScore**, a lower-is-better count of voice-leading violations below.

i. parallel fifths
ii. parallel octaves
iii. voice crossings
iv. hidden fifths (outer voices)
v. hidden octaves (outer voices)
vi. spacing violations
vii. large leap (> an octave)
viii. augmented leap

Each rule contributes one violation count per occurrence; the checker runs over rendered MIDI.

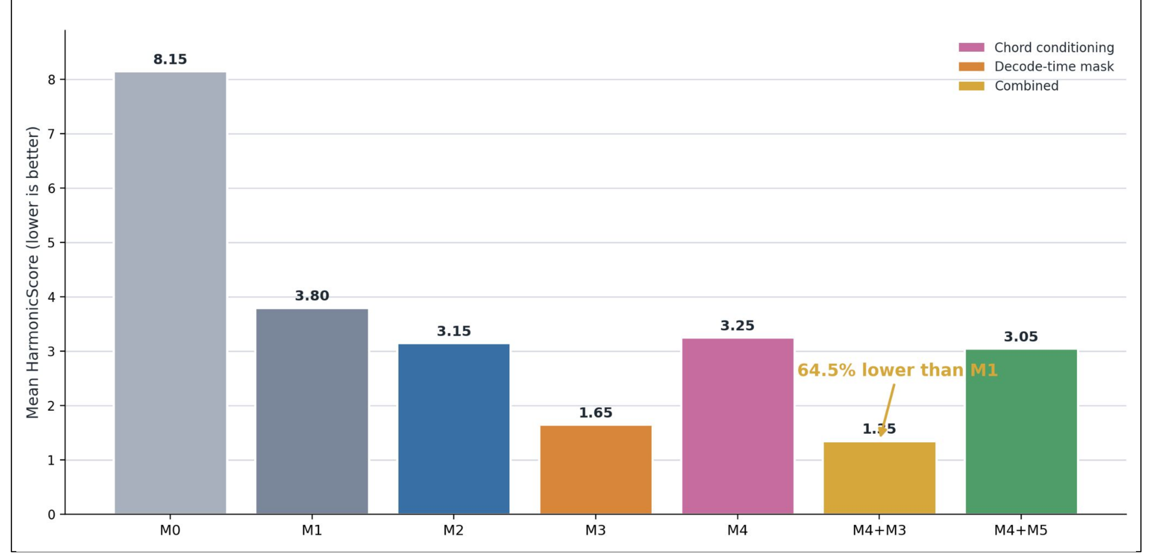
Quantitative Result

HarmonicScore = Σ *k=1…8* violations

n = 20 generated pieces per arm

Lower is better. Zero would mean a chorale free of textbook violations — which not even Bach achieves on every spot-check.

We run the rule checker on all models including SOTA (denoted as M0). The result shows that **chord conditioning alone helps modestly; rule-masked decoding is strongest, and the combination performs best.**



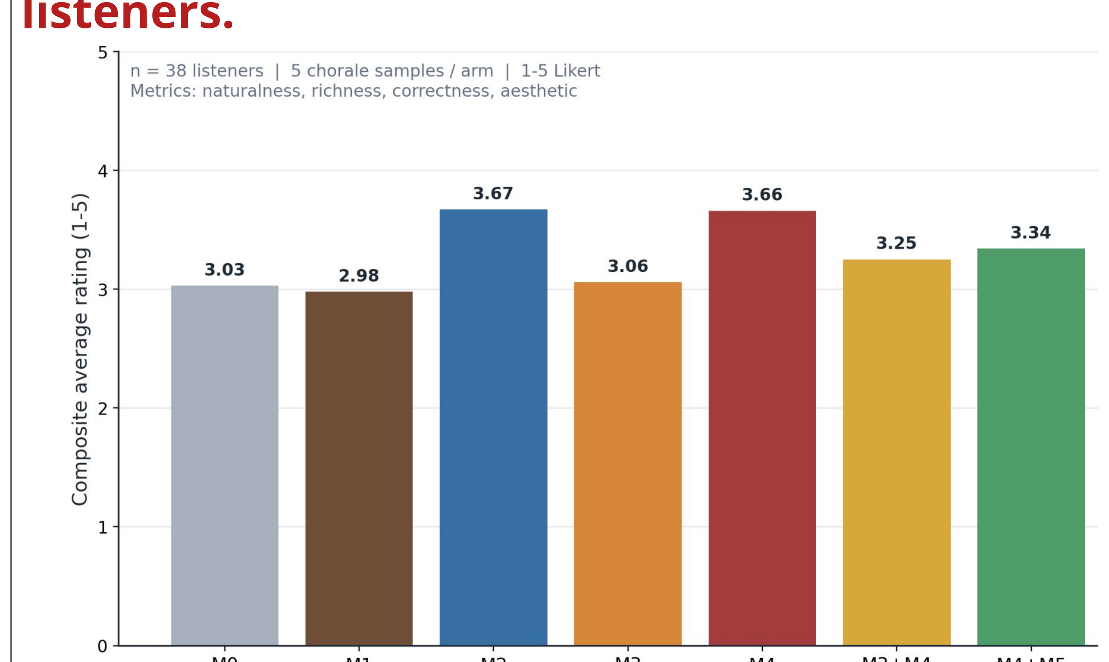
Qualitative Result

Participants: N = 38 listeners (24% formal training, 56% self-taught / hobbyist, 21% no formal training, 70% at least somewhat familiar with Bach)

Survey:

Score 1) Naturalness, 2) Richness, 3) Correctness, 4) Aesthetic from 1 to 5 (5 being the best).

The twist — **best on HarmonicScore ≠ best to listeners.**



Discussions

Key Findings.

Performance vs. Perception: Both objective metrics and human surveys confirm that rule-injected models (M2–M4) significantly outperform the baseline (M0 & M1).

Top Performers: M2 and M4 achieved the highest composite human ratings, while combining training-time conditioning (M4) with decode-time enforcement (M3) yielded the lowest rule violation result by 64.5%.

Most Effective Stage: M4's success demonstrates that combining training-time conditioning with decode-time enforcement creates the most "correct" and "pleasing" results.

Limitations.

Correctness vs Musicality: Although M4 achieved a 64.5% reduction in formal violations, M2 maintained a marginal lead in human preference despite lower rule compliance. This indicates that rigid adherence to voice-leading syntax does not linearly correlate with perceived musicality.

Evaluation Scope. HarmonicScore prioritizes local voice-leading syntax and may lack the sensitivity to evaluate long-range melodic phrasing or the higher-level structural intentions found in original compositions.

Future Work.

Our future research aims to implement differentiable "soft" constraints for expressive rule-breaking while integrating expert qualitative feedback to bridge the gap between algorithmic correctness and human artistry.

Works Cited

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JSB chorale dataset, music21 (for Roman-numeral analysis), pretty_midi (for MIDI I/O), PyTorch (for training).